at $x=4$ it is not continuous, at $x=11$, tangent is perpendicular

(6) does not exist

(7) (i) not differentiable at $x=n\pi, n\in z$ (ii) not differentiable at $x=(2n+1)\frac{\pi}{2}, n\in z$

Exercise 10.2

(1) $1-3\cos x$ (2) $\cos x-\sin x$ (3) $x\cos x+\sin x$

(4) $-\sin x-2\sec^2 x$ (5) $3t^2\cos t-t^3\sin t$ (6) $4\sec t\tan t+\sec^2 t$

(7) $e^x(\cos x+\sin x)$ (8) $\frac{x\sec^2 x-\tan x}{x^2}$ (9) $\frac{1}{1+\cos x}$

(10) $\frac{(1-x)\cos x+(1+x)\sin x}{(\sin x+\cos x)^2}$ (11) $\cos x+\sin x$ (12) $\frac{x\cos x-2\sin x}{x^3}$

(13) $\tan\theta\sec\theta+\cos\theta+\sin\theta$ (14) $-\frac{(1+\cos^2 x)}{\sin^3 x}$ (15) $x\cos 2x+\sin x\cos x$

(16) $e^{-x}\left[\frac{1}{x}-\log x\right]$ (17) $e^{-3x}\left[-3(x^2+5)\log(1+x)+\frac{x^2+5}{1+x}+2x\log(1+x)\right]$

(18) $\frac{\pi}{180}\cos\frac{\pi}{180}x$ (19) $\frac{\log_{10} e}{x}$

Exercise 10.3

(1) $5(2x+4)(x^2+4x+6)^4$ (2) $3\sec^2 3x$ (3) $-\sec^2 x\sin(\tan x)$

(4) $x^2(1+x^3)^{-\frac{2}{3}}$ (5) $\frac{1}{2\sqrt{x}}e^{\sqrt{x}}$ (6) $e^x\cos(e^x)$

(7) $7(3x^2+4)(x^3+4x)^6$ (8) $\frac{3}{2}\left(t-\frac{1}{t}\right)^{\frac{1}{2}}\left(1+\frac{1}{t^2}\right)$ (9) $\frac{1}{3}\sec^2 t(1+\tan t)^{-\frac{2}{3}}$

(10) $-3x^2\sin(a^3+x^3)$ (11) $-my$ (12) $20\sec 5x\tan 5x$

(13) $\frac{8(2x-5)^3}{(8x^2-5)^4}[-4x^2+30x-5]$ (14) $\frac{8x^3+14x}{3(x^2+2)^{\frac{2}{3}}}$ (15) $e^{-x^2}[1-2x^2]$

(16) $-\frac{3t^2}{2(t^3+1)^{\frac{3}{4}}(t^3-1)^{\frac{5}{4}}}$ (17) $\frac{14-3x}{2(7-3x)\sqrt{7-3x}}$ (18) $-\sin x\sec^2(\cos x)$

(19) $\sin x(1+\sec^2 x)$ (20) $\frac{5^{-\frac{1}{x}}(\log 5)}{x^2}$ (21) $\frac{\sec^2 x}{\sqrt{1+2\tan x}}$

(22) $3\sin x\cos x(\sin x-\cos x)$ (23) $-k\sin kx\sin(2\cos kx)$ (24) $-6\sin 2x(1+\cos^2 x)^5$

(25) $\frac{3e^{3x}+2e^{4x}}{(1+e^x)^2}$ (26) $\frac{2\sqrt{x}+1}{4\sqrt{x}\sqrt{x+\sqrt{x}}}$ (27) $e^{x\cos x}[\cos x-x\sin x]$

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